

CareMap Facility Analytics

Project Report

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Abstract

This report documents the preparation and analysis of a historical national health-centre directory containing 200,438 records. I standardised facility types and geography labels, validated coordinates, measured field completeness, removed 6,655 exact duplicate signatures, and created a 193,783-row analytical table.

Five experiments examine data cleaning, facility type distribution, state concentration, rural and urban labels, and quality indicators. The project supplies a four-page Power BI design while clearly separating historical directory coverage from current healthcare capacity or access.

Keywords: health facility directory; data quality; geospatial analysis; Power BI; public health data

Report structure

The report presents the problem, source data, preparation method, evaluation design, five evidence-based experiments, limitations, reproducibility details, and conclusion.

1. Project overview and problem statement

CareMap turns a historical national health-centre directory into a governed facility-level analytical table and Power BI implementation plan. I built it because a large directory can still contain duplicates, invalid coordinates, old geography labels, and inconsistent facility types that distort maps and counts.

The analytical purpose is to compare the distribution and quality of published facility records while keeping the source date visible. The result is not a current service locator and does not measure staffing, beds, medicine availability, clinical quality, or patient access.

2. Source and preparation method

The project downloads the All India Health Centres Directory from Kaggle. Its description attributes the data to data.gov.in and dates the snapshot to 7 October 2016. That historical date is a central interpretation constraint.

I standardised nulls and five facility categories, merged an outdated Andhra Pradesh label, validated coordinates against a broad India range, calculated field completeness, and removed exact duplicate signatures using location and facility attributes.

Measure	Result
Raw records	200,438
Clean records	193,783
Duplicates removed	6,655
State and union territory labels	36
State-district combinations	688

3. Analytical and Power BI design

The clean table preserves facility name, type, ownership, location type, state, district, subdistrict, address, coordinates, activity fields, and record completeness. Reusable DAX measures calculate facility count, geography coverage, mapped percentage, public-label percentage, and average completeness.

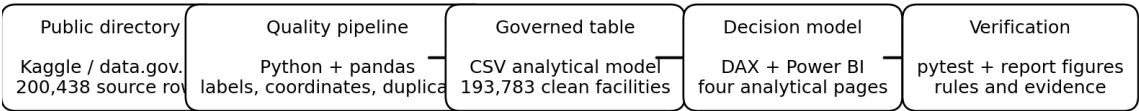
The four Power BI pages cover national overview, geographic exploration, state comparison, and data quality. Counts are always described as directory records rather than current capacity.

4. End-to-end analytical architecture

The architecture separates source preservation, transformation, governed output, decision modelling and verification. The acquisition stage records source provenance before Python and pandas standardise categories, validate coordinates and calculate duplicate signatures.

The clean facility table is the only aggregation input used by the DAX measures and Power BI page design. Pytest and generated evidence figures verify the transformation rules independently from the dashboard. This separation prevents a visual page from silently applying a second cleaning rule.

CareMap end-to-end analytical architecture



Architecture diagram. Technology and responsibility across the CareMap data flow.

Stage design: Each box has one primary input, technology and output so the source, pipeline and dashboard responsibilities remain distinct.

Control point: The governed facility table is created before DAX or visual aggregation, which keeps record counts consistent across pages.

Result: The same 193,783-row table supports the quality summary, analytical figures and Power BI implementation plan.

5. Automated pipeline test execution

I executed the repository test suite after the report and filename changes. Four tests passed. The suite checks required columns, coordinate validation, duplicate removal and standardised facility categories.

These tests do not prove that every historical source value is correct. They verify that the declared preparation rules behave consistently and that invalid or repeated records follow the documented path.

```
Actual pipeline test execution

$ .venv/bin/pytest -q
tests/test_pipeline.py .... [100%]

4 passed in 0.42s

Validated: required columns, coordinate rules,
duplicate removal and governed facility categories.
```

Test execution evidence. Actual CareMap pytest result captured for this report revision.

Execution: The command completed with four passing tests and no failed test.

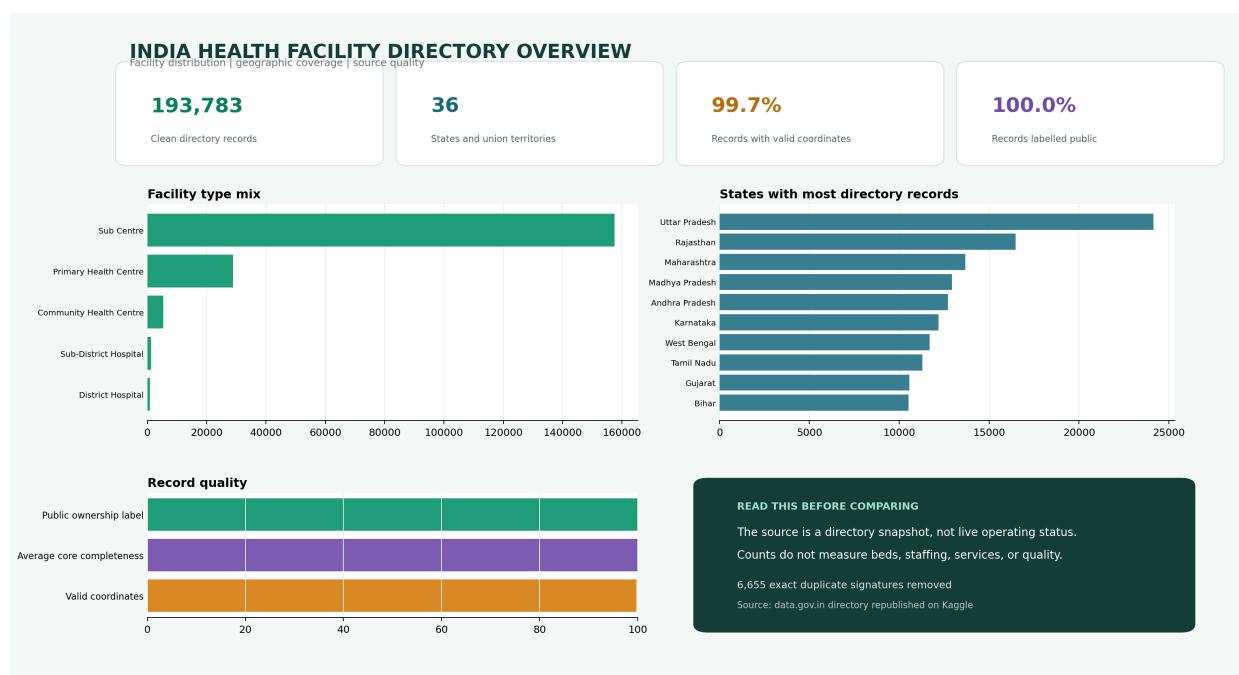
Coverage: The checks focus on transformation rules that directly affect counts, maps and category comparisons.

Boundary: Source freshness and independent geographic verification remain outside automated unit-test coverage.

6. Dashboard evidence and decision use

The dashboard evidence brings the governed totals, facility mix, geography and data-quality indicators into one decision view. It is included to show how the prepared table is consumed, not to replace the reproducible DAX and build guide.

A user can start from the national count, filter to a state or facility type, and keep mapped percentage and completeness visible while interpreting the result. This reduces the risk of treating a large record count as a complete measure of current healthcare access.



Dashboard evidence. CareMap analytical view generated from the clean facility table.

Question: Can the dashboard expose scale, composition and data quality at the same time?

Observed view: The evidence combines governed KPIs with facility and geographic breakdowns rather than presenting one unsupported total.

Use: The page supports exploratory comparison while the report retains the source-date and interpretation limitations.

7. Experiment 1: cleaning results

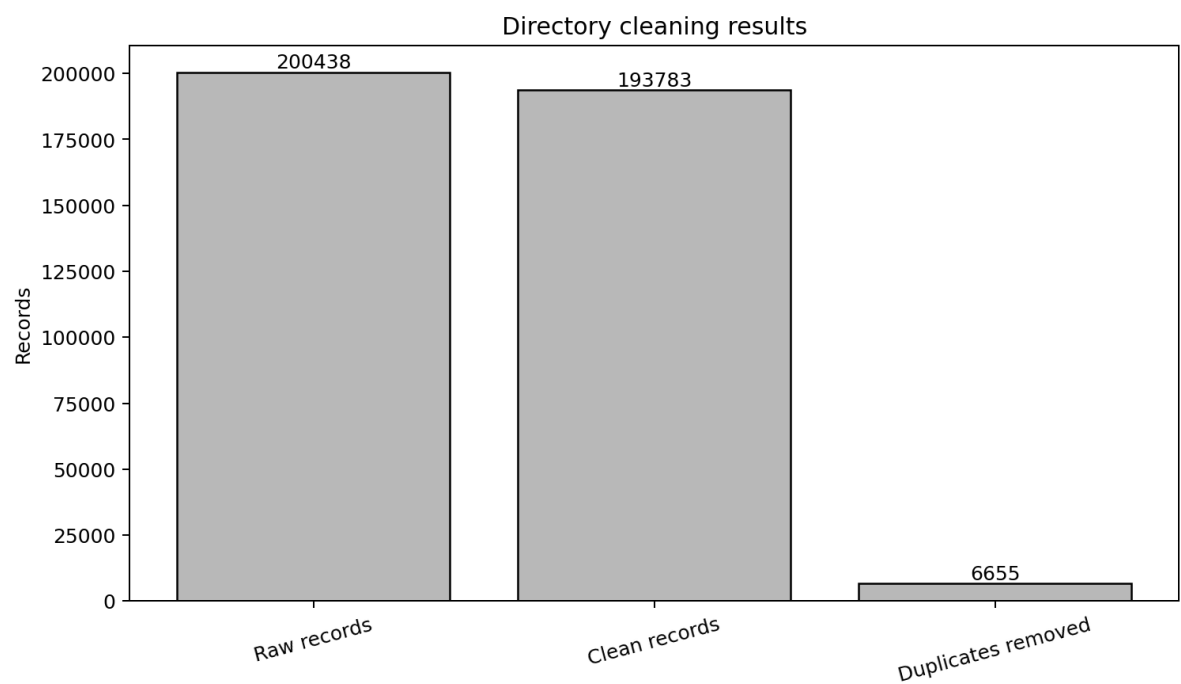


Figure 1. Raw records, clean records, and removed duplicate signatures.

What I tested: Whether the signature rule identifies exact repeated facilities without collapsing similarly named facilities in different places.

What the graph shows: The pipeline removes 6,655 duplicates and retains 193,783 clean rows.

Conclusion: Deduplication changes the national total materially and must occur before dashboard aggregation.

8. Experiment 2: facility type distribution

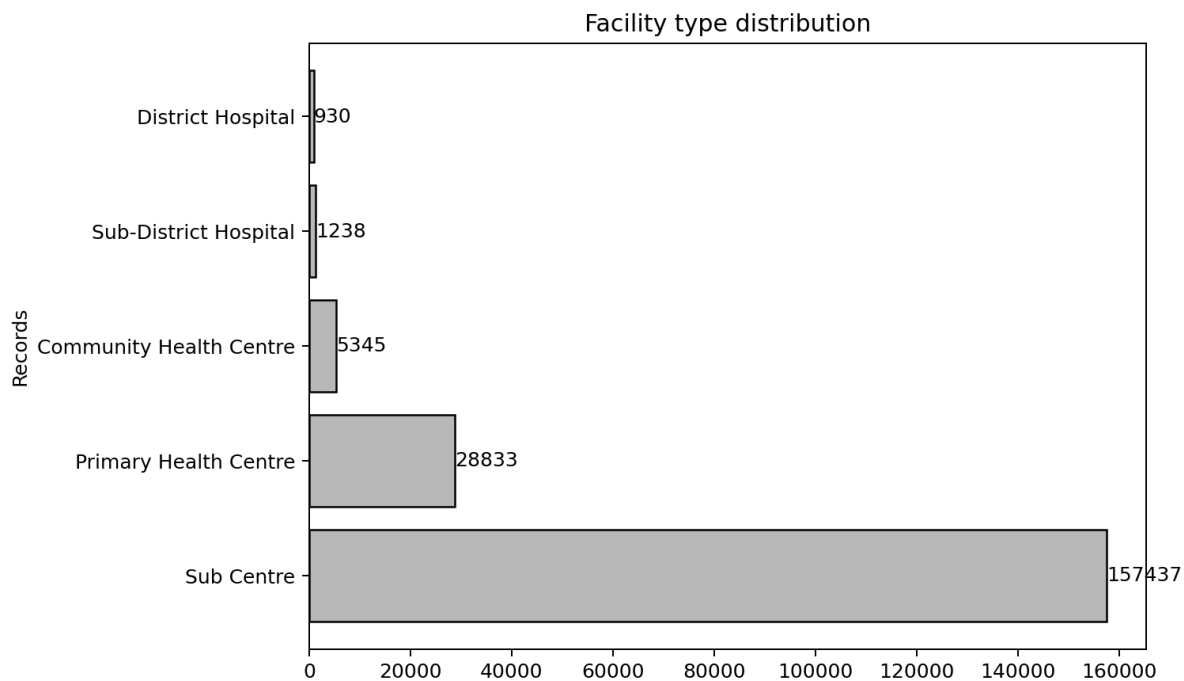


Figure 2. Clean directory records by standardised facility type.

What I tested: How the five governed facility categories contribute to the national directory.

What the graph shows: Sub Centres dominate, followed by Primary Health Centres; hospitals form a much smaller share.

Conclusion: State totals mainly describe lower-level facilities, so facility type must remain visible in comparisons.

9. Experiment 3: state distribution

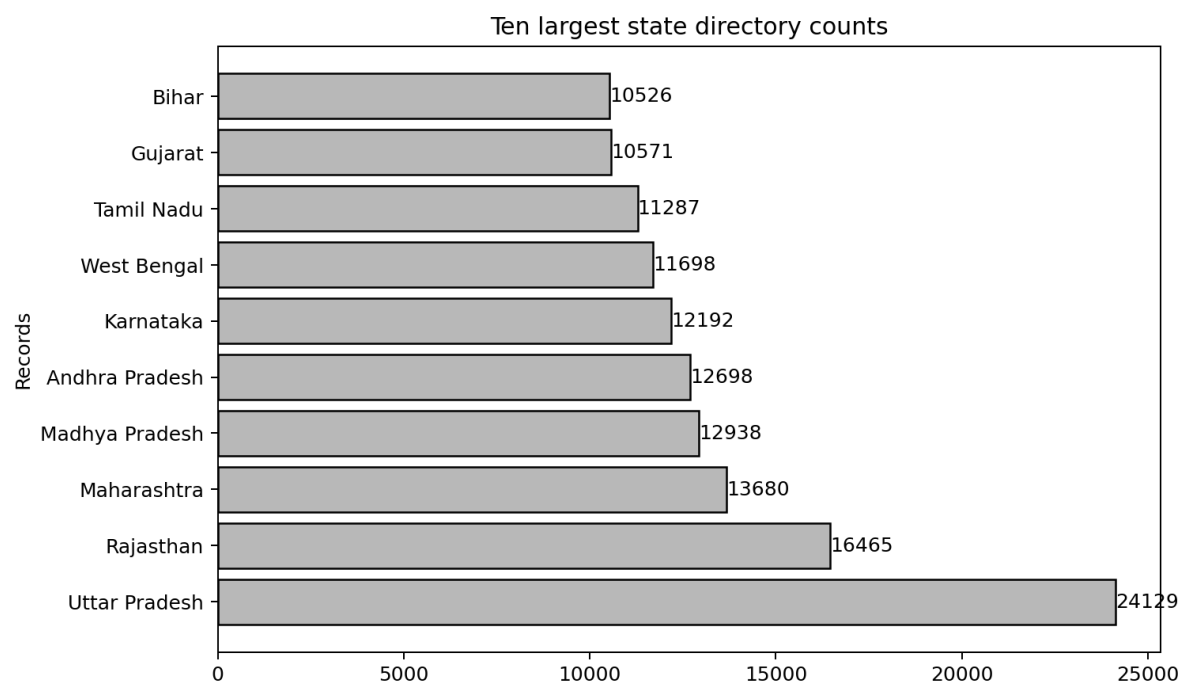


Figure 3. Ten largest state directory counts in the historical snapshot.

What I tested: Whether directory records are evenly distributed across states.

What the graph shows: Uttar Pradesh has the largest count, followed by Rajasthan and Maharashtra.

Conclusion: The ranking describes source coverage and administrative scale, not health need or per-person access.

10. Experiment 4: rural and urban mix

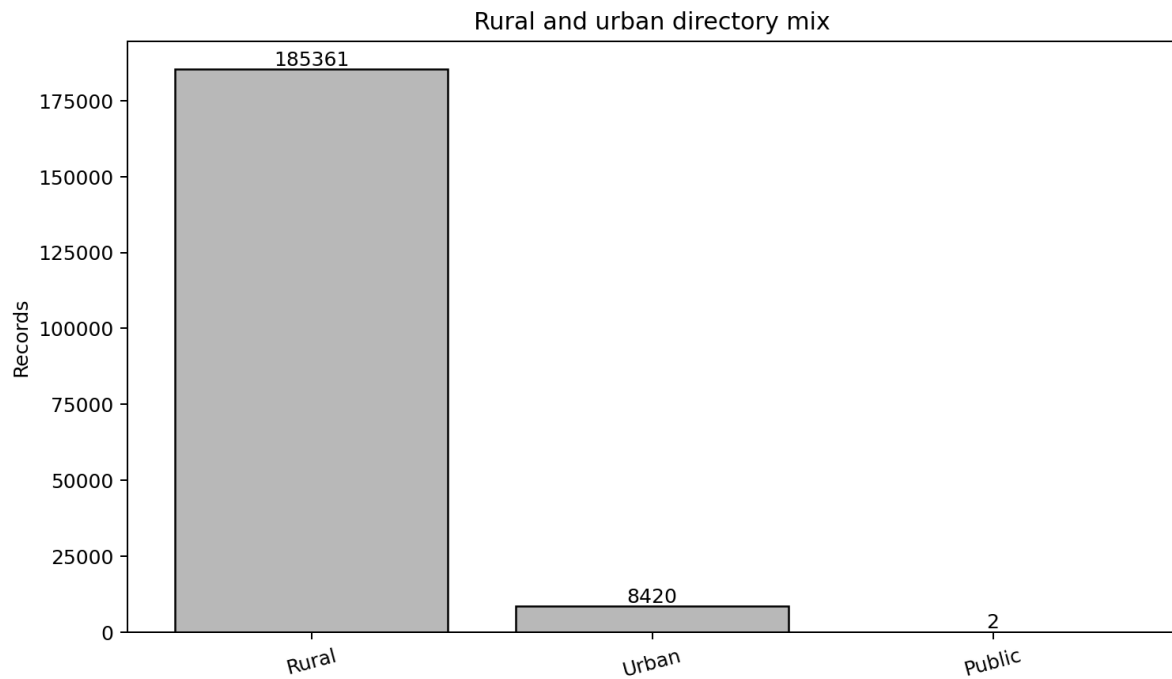


Figure 4. Directory records by source location-type label.

What I tested: How the source labels facilities across rural and urban settings.

What the graph shows: The directory is overwhelmingly rural, with only 8,420 urban records and two unusual Public labels.

Conclusion: The location field requires governance, and the dataset is not balanced for general urban-versus-rural capacity claims.

11. Experiment 5: data quality indicators

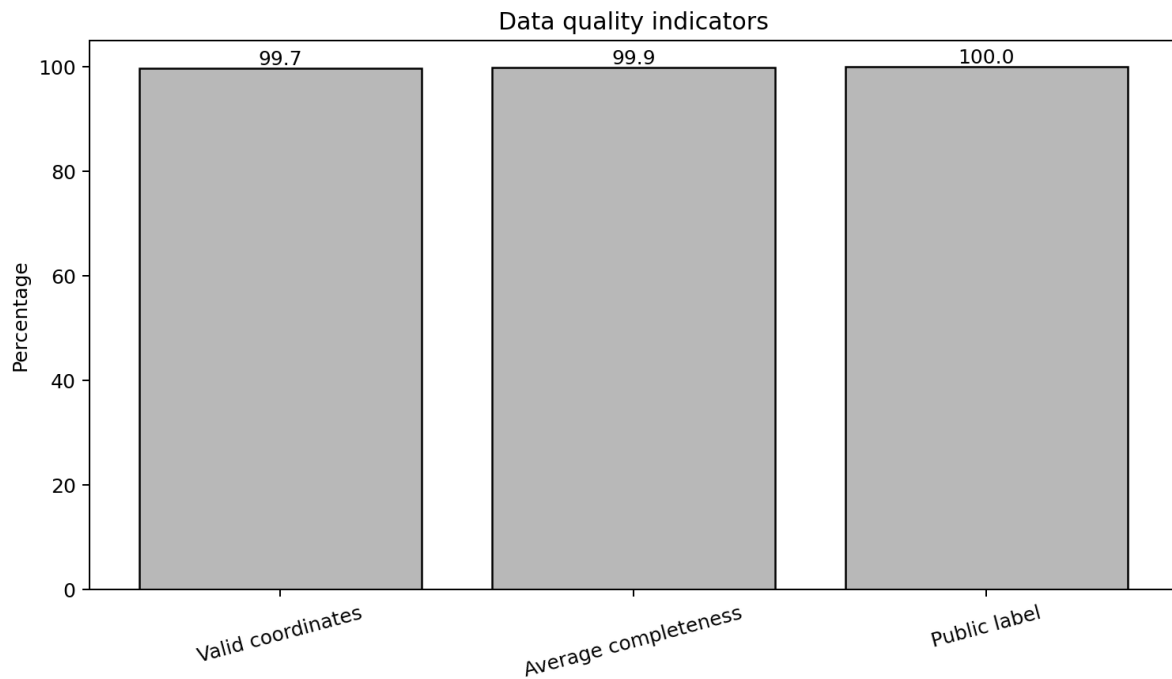


Figure 5. Coordinate, completeness, and public-label quality measures.

What I tested: Whether the clean records contain the core fields required for mapping and segmentation.

What the graph shows: Valid coordinates are 99.7 percent and average completeness is 99.9 percent.

Conclusion: Field presence is strong, but it does not prove that historical coordinates or ownership labels remain current.

12. Results and interpretation

The final table contains 193,783 records across 36 state and union territory labels and 688 state-district combinations. 193,151 rows contain coordinates inside the broad validation range.

The largest facility type is Sub Centre. Uttar Pradesh has the largest directory count. These findings describe a 2016 snapshot and should not be converted into statements about current service quality, utilisation, staffing, or population access.

13. Limitations and reproducibility

The dataset is old, largely public-facility focused, and lacks current verification, population denominators, road networks, travel time, beds, staff, services, and outcomes. Coordinate presence is not independent location validation.

The repository versions the acquisition, cleaning pipeline, quality summary, dashboard generator, DAX, tests, five figures, and this report. A future study should obtain a current authoritative register, validate a geographic sample, join compatible population data, and calculate travel-time coverage for a carefully defined service level.

14. Conclusion

CareMap demonstrates that facility analytics must begin with source age, duplicates, geography labels, coordinate rules, and interpretation boundaries. I produced a clean national table, auditable quality metrics, five evidence figures, and a Power BI plan while avoiding the unsupported claim that a historical directory measures present healthcare access.